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OPTIMIZATION

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1 Linear Programing

Linear Inequality: For any linear function $f(x_1, x_2....x_n)$ and any real number b the linear inequality can be expressed as:

$$f(x_1, x_2....x_n) \geq b$$

Example: $2x_1 + 3x_2 \leq 3$

Def 1.1: A linear Programing problem is an optimization problem for which we do:

1. Maximize the Profit
2. Minimize the Cost

(a) **Objective Funtion:** Creating a function using decision variables. *example:*

$$2x_1 + 3x_2 \leq 5$$

(b) **Decicion Variables:** The parameters we want to maximize of minimize according to the given constraints.

NB: Decision variable $\propto$ Objective function

- *Problem Solving Algorithm:-*

1. Identify the decision variables;
2. Create a Objective function;
3. Apply the constraints.

example: Farmer Jones must determine how many acres of corn and wheat to plant this year. An acre of wheat yields 25 bushels of wheat and requires 10 hours of labor per week. An acre of corn yields 10 bushels of corn and requires 4 hours of labor per week. All wheat can be sold at Tk4 a bushel, and all corn can be sold at Tk3 a bushel. Seven acres of land and 40 hours per week of labor are available. Government regulations require that at least 30 bushels of corn be produced during the current year. formulate an LP whose solution will tell Farmer Jones how to maximize the total revenue from wheat and corn.

Solution: Let,

x_1=Number of bushels of wheat;
x_2=Number of bushels of corn;

So the Objective function will be,

$$z = 4x_1 + 3x_2$$

Now the constraints are,

$$\begin{gathered}\frac{10x_1}{25} + \frac{4x_2}{10} \leq 40 \\ \frac{1x_1}{25} + \frac{1x_2}{10} \leq 7 \\ x_2 \geq 30 \\ x_1 \geq 0 \\ x_2 \geq 0\end{gathered}$$

•***Matrix Represenation of LP:-***

$$\begin{gathered}\text{maximize or minimize } z = \underline{c}\underline{x}^T \\ \text{s.t } A\underline{x} \geq (\leq)\underline{b} \text{ and} \\ x \geq 0\end{gathered}$$

$$\text{here } \underline{c} = \begin{pmatrix} c_1 \\ c_2 \\ . \\ . \\ c_n \end{pmatrix} \quad \text{and} \quad \underline{b} = \begin{pmatrix} b_1 \\ b_2 \\ . \\ . \\ b_n \end{pmatrix}$$

$$\underline{x} = \begin{pmatrix} x_1 \\ x_2 \\ . \\ . \\ x_n \end{pmatrix}$$

2 Geometric preliminaries and solutions

• An LP with only two variables can be solved graphically

Convex set: A subset k in $\mathbb{R}^n$ is a convex set if k is zero or k is a single point or for each two distinct points p and q in k the line segment connecting p and q entirely lies in k.

1. Convex set can be bounded
2. Convex set can be unbounded

Feasible Region: A Region which satisfies all the condition is a feasible region. Feasible region is necessarily convex.

Half Space: A half space in $\mathbb{R}^n$ is the set of all points which satisfies the inequality:-

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n \leq c$$

or the inequality:-

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n \geq c$$

NB:Linear Inequality=Half Space

Hyper-plane: A hyperplane in $\mathbb{R}^n$ is the all sets that satisfies:-

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = c$$

Convex combination: If $\underline{p} = (p_1, p_2, \cdots, p_n)$ and $\underline{q} = (q_1, q_2, \cdots, q_n)$ are points in $\mathbb{R}^n$ then the line segment joining $\underline{p}$ and $\underline{q}$ consists of all points of the form:-

$$(1-t)\underline{p} + t\underline{q}; \qquad t \in [-1, 1]$$

• Mathematically a set k$\subset \mathbb{R}^n$ is a convex set if for any $p, q \in k$

1. $(1-t)p + tq \in k \qquad \forall\ t \in [0, 1]$
2. $(1-s)p + sq \in k \qquad \forall\ s \in [0, 1]$

***THEOREM* 2.1.** *Every half space is a convex set*

Proof. Take any 2 set of point $\underline{p_1}$ and $\underline{p_2}$ from H.
consider the half space $\underline{a}^T\underline{x} \leq c$

$$\begin{aligned} \underline{a}^T\underline{p_1} &\leq c \\ \underline{a}^T\underline{p_2} &\leq c \end{aligned} \tag{1}$$

Now take, $\underline{p} : t\underline{p_1} + (1-t)\underline{p_2} \qquad t \in [0,1]$

$$\begin{aligned} &\textit{Therefore} && \underline{a}^T\underline{p} = \underline{a}^T[t\underline{p_1} + (1-t)\underline{p_2}] \\ \implies && & \underline{a}^T\underline{p} = \underline{a}^T t\underline{p_1} + \underline{a}^T(1-t)\underline{p_2}] \\ \implies && & \underline{a}^T\underline{p} \leq tc + (1-t)c \\ \implies && & \underline{a}^T\underline{p} \leq c \\ \implies && & \underline{a}^T(t\underline{p_1} + (1-t)\underline{p_2} \leq c \qquad t \in [0,1] \end{aligned}$$

and $t\underline{p_1} + (1-t)\underline{p_2} \in$ H . Hence H is a convex set

□

- **Identifying the side of hyperplane where the halfspace lies**:

We have to draw the hyperplane first (according to the ojective function). Then we need to verify in which sodes of the hyperplane satisfies all of the condition and constrainsts of the optimization problem.
To verify this we have 2 ways:-

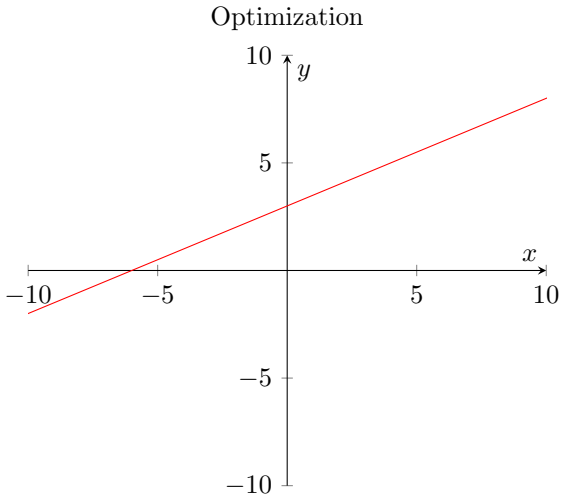
1. The Gradient(∇) vector:- Calculate the gradient vector and its direction.
2. Point method:- Pick a point and put it in the objective function to verify.

example:- $-2x_1 + 4x_2 \geq 12$ verify the half space.

Solution: Standerdizing the given equation:

$$\begin{aligned} & -2x_1 + 4x_2 \geq 12 \\ \implies & \frac{-x_1}{6} + \frac{x_2}{3} = 1 \end{aligned}$$

Now drawing the figure:-



Now substituting $(0, 0)$ in the equation we get:-

$$-2(0) + 4(0) = 0$$

Which is less then 12 not greater than 12. So the half space will be in the opposite side of the line where $(0, 0)$ is.

- Now using the gradint vector:-

$$\frac{\partial f}{\partial x_1} + \frac{\partial f}{\partial x2} =< -2, 4 >$$

This function grows rapidly in the direction of $(-2, 4)$. Therefore the half space will be in the side of the point (-2,4).